```python
from pathlib import Path
import json

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

from IPython.display import display, Markdown

plt.rcParams["figure.dpi"] = 130
pd.set_option("display.max_colwidth", 200)
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

1. Locate and validate the saved artefacts

```python
# Change this if your outputs live elsewhere.
OUTDIR = Path("./analysis_outputs")

required_files = [
    "xgb_analysis_train.csv",
    "xgb_deciles.csv",
    "xgb_mean_abs_shap.csv",
    "xgb_mean_signed_shap.csv",
    "xgb_regime_signed_shap.csv",
    "xgb_case_type_signed_shap.csv",
    "xgb_interaction_candidates.csv",
    "xgb_xai_summary.json",
    "realmlp_analysis_train.csv",
    "realmlp_analysis_test.csv",
    "realmlp_fold_metrics.csv",
    "realmlp_schema_check.csv",
    "realmlp_deciles.csv",
    "realmlp_permutation_importance.csv",
    "realmlp_finite_difference_sensitivity.csv",
    "realmlp_grouped_ablation.csv",
    "realmlp_uncertainty_summary.csv",
    "realmlp_xai_summary.json",
]

missing = [f for f in required_files if not (OUTDIR / f).exists()]
if missing:
    raise FileNotFoundError(
```

```
        "The following expected artefacts were not found in "
        f"{OUTDIR.resolve()}:\n- " + "\n- ".join(missing)
    )

print("Using output directory:", OUTDIR.resolve())
print("All required exported tables and summaries were found.")
```

```
Using output directory: /Users/abhishekdey/Downloads/Predicting-Churn/analysis_outputs
All required exported tables and summaries were found.
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

2. Load exported tables and summaries

```
xgb_train = pd.read_csv(OUTDIR / "xgb_analysis_train.csv")
xgb_dec = pd.read_csv(OUTDIR / "xgb_deciles.csv")
xgb_mean_abs_shap = pd.read_csv(OUTDIR / "xgb_mean_abs_shap.csv")
xgb_mean_signed_shap = pd.read_csv(OUTDIR /
"xgb_mean_signed_shap.csv")
xgb_regime_signed = pd.read_csv(OUTDIR / "xgb_regime_signed_shap.csv",
index_col=0)
xgb_case_type_signed = pd.read_csv(OUTDIR /
"xgb_case_type_signed_shap.csv", index_col=0)
xgb_interactions = pd.read_csv(OUTDIR /
"xgb_interaction_candidates.csv")
xgb_summary = json.loads((OUTDIR /
"xgb_xai_summary.json").read_text())

real_train = pd.read_csv(OUTDIR / "realmlp_analysis_train.csv")
real_test = pd.read_csv(OUTDIR / "realmlp_analysis_test.csv")
real_fold_metrics = pd.read_csv(OUTDIR / "realmlp_fold_metrics.csv")
real_schema = pd.read_csv(OUTDIR / "realmlp_schema_check.csv")
real_dec = pd.read_csv(OUTDIR / "realmlp_deciles.csv")
real_perm = pd.read_csv(OUTDIR / "realmlp_permutation_importance.csv")
real_sens = pd.read_csv(OUTDIR /
"realmlp_finite_difference_sensitivity.csv")
real_ablation = pd.read_csv(OUTDIR / "realmlp_grouped_ablation.csv")
real_unc = pd.read_csv(OUTDIR / "realmlp_uncertainty_summary.csv",
index_col=0)
real_summary = json.loads((OUTDIR /
"realmlp_xai_summary.json").read_text())

# normalise one XGBoost export depending on whether it came out as
```

```
unnamed+value or named columns
if xgb_mean_abs_shap.shape[1] == 2 and "mean_abs_shap" in
xgb_mean_abs_shap.columns:
    feature_col = [c for c in xgb_mean_abs_shap.columns if c !=
"mean_abs_shap"][0]
    xgb_mean_abs_shap = xgb_mean_abs_shap.rename(columns={feature_col:
"feature"})
if xgb_mean_signed_shap.shape[1] == 2 and "mean_signed_shap" in
xgb_mean_signed_shap.columns:
    feature_col = [c for c in xgb_mean_signed_shap.columns if c !=
"mean_signed_shap"][0]
    xgb_mean_signed_shap =
xgb_mean_signed_shap.rename(columns={feature_col: "feature"})

print("Loaded all exported outputs.")
print("XGB train rows:", len(xgb_train))
print("RealMLP train rows:", len(real_train))
```

```
Loaded all exported outputs.
XGB train rows: 594194
RealMLP train rows: 594194
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

3. High-level comparison table

```
def _safe_float(d, key):
    val = d.get(key, np.nan)
    try:
        return float(val)
    except Exception:
        return np.nan

comparison_summary = pd.DataFrame([
    {
        "model": "XGBoost",
        "auc": _safe_float(xgb_summary, "train_auc_competition_only"),
        "logloss": _safe_float(xgb_summary,
"train_logloss_competition_only"),
        "brier": _safe_float(xgb_summary,
"train_brier_competition_only"),
        "n_features_runtime": xgb_summary.get("n_runtime_cols",
np.nan),
        "n_feature_importance_table_rows": len(xgb_mean_abs_shap),
```

```
    },
    {
        "model": "RealMLP",
        "auc": _safe_float(real_summary, "oof_auc"),
        "logloss": _safe_float(real_summary, "oof_logloss"),
        "brier": _safe_float(real_summary, "oof_brier"),
        "n_features_runtime": real_summary.get("n_numeric_features",
np.nan),
        "n_feature_importance_table_rows": len(real_perm),
    },
]).sort_values("auc", ascending=False).reset_index(drop=True)

display(comparison_summary)
```

```
     model       auc   logloss     brier  n_features_runtime  \
0  XGBoost  0.933437  0.268055  0.085993                 161
1  RealMLP  0.917199  0.298696  0.095583                  22

   n_feature_importance_table_rows
0                              161
1                               22
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

Use this table to anchor the report quantitatively. The point is not only which model has the larger AUC, but also whether the gains are accompanied by better log-loss and Brier score. If one model is stronger on discrimination but not on probability quality, say that explicitly.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

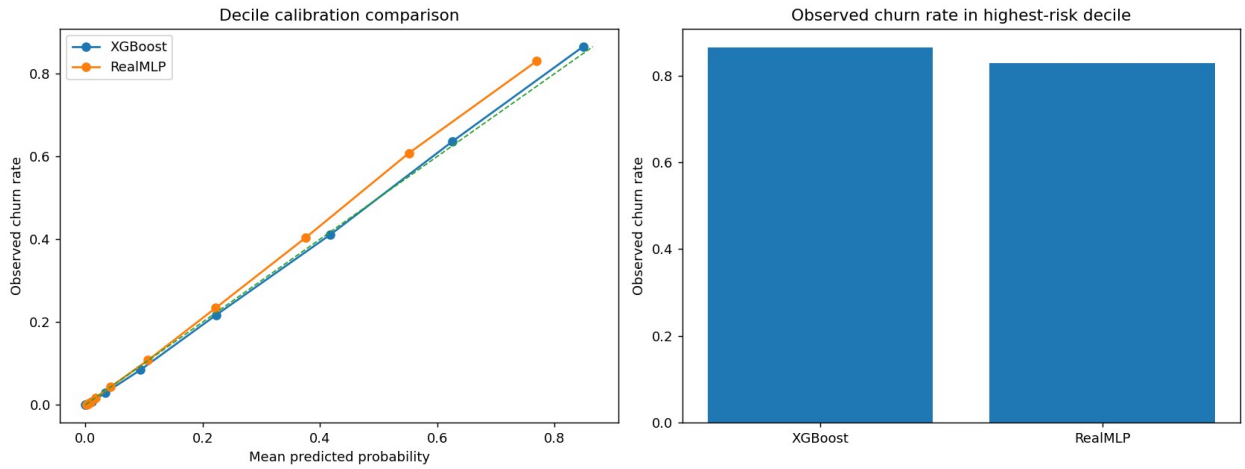
4. Calibration and risk concentration comparison

```
# make the decile tables easy to compare
xgb_dec_cmp = xgb_dec.rename(columns={
    "mean_pred": "xgb_mean_pred",
    "churn_rate": "xgb_observed_rate",
    "n": "xgb_n",
    "pred_min": "xgb_pred_min",
    "pred_max": "xgb_pred_max",
```

```
})
real_dec_cmp = real_dec.rename(columns={
    "mean_pred": "real_mean_pred",
    "churn_rate": "real_observed_rate",
    "n": "real_n",
    "pred_min": "real_pred_min",
    "pred_max": "real_pred_max",
})

dec_compare = xgb_dec_cmp.merge(real_dec_cmp, on="bin", how="inner")
display(dec_compare)
```

```
   bin  xgb_mean_pred  xgb_observed_rate  xgb_n  xgb_pred_min
xgb_pred_max  \
0    0       0.000052           0.000000  59420      0.000003
0.000110
1    1       0.000682           0.000236  59419      0.000110
0.001696
2    2       0.003563           0.002003  59419      0.001696
0.005920
3    3       0.011040           0.007927  59420      0.005920
0.018740
4    4       0.033567           0.028728  59419      0.018741
0.053924
5    5       0.093750           0.084165  59419      0.053925
0.145461
6    6       0.223051           0.217115  59420      0.145461
0.313004
7    7       0.417723           0.411215  59419      0.313006
0.523134
8    8       0.625240           0.635672  59419      0.523141
0.727878
9    9       0.848388           0.865012  59420      0.727878
0.999883

   real_mean_pred  real_observed_rate  real_n  real_pred_min
real_pred_max
0        0.002330            0.001094   59420       0.000880
0.003107
1        0.004047            0.002710   59419       0.003107
0.005285
2        0.007776            0.005840   59419       0.005285
0.011012
3        0.017026            0.016829   59420       0.011013
0.025836
4        0.042658            0.042966   59419       0.025837
0.065962
5        0.106499            0.107390   59419       0.065962
0.156202
6        0.221997            0.233793   59420       0.156208
```

```
0.294355
7        0.375918            0.404180   59419       0.294356
0.460177
8        0.551160            0.607600   59419       0.460180
0.647994
9        0.768880            0.829670   59420       0.648000
0.966818


fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# calibration-like line comparison
axes[0].plot(dec_compare["xgb_mean_pred"],
dec_compare["xgb_observed_rate"], marker="o", label="XGBoost")
axes[0].plot(dec_compare["real_mean_pred"],
dec_compare["real_observed_rate"], marker="o", label="RealMLP")
diag_min =
min(dec_compare[["xgb_mean_pred","xgb_observed_rate","real_mean_pred",
"real_observed_rate"]].min())
diag_max =
max(dec_compare[["xgb_mean_pred","xgb_observed_rate","real_mean_pred",
"real_observed_rate"]].max())
axes[0].plot([diag_min, diag_max], [diag_min, diag_max],
linestyle="--", linewidth=1)
axes[0].set_title("Decile calibration comparison")
axes[0].set_xlabel("Mean predicted probability")
axes[0].set_ylabel("Observed churn rate")
axes[0].legend()

# top-decile lift / concentration comparison
lift_df = pd.DataFrame({
    "model": ["XGBoost", "RealMLP"],
    "top_decile_observed_rate": [
        dec_compare["xgb_observed_rate"].iloc[-1],
        dec_compare["real_observed_rate"].iloc[-1],
    ],
    "bottom_decile_observed_rate": [
        dec_compare["xgb_observed_rate"].iloc[0],
        dec_compare["real_observed_rate"].iloc[0],
    ],
})
axes[1].bar(lift_df["model"], lift_df["top_decile_observed_rate"])
axes[1].set_title("Observed churn rate in highest-risk decile")
axes[1].set_ylabel("Observed churn rate")

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are cross-model plots and should be compared directly. In the left panel, the closer a model stays to the diagonal, the better its decile-level probability calibration. In the right panel, the higher the top-decile observed rate, the better the model is at concentrating true churners into the most actionable segment.

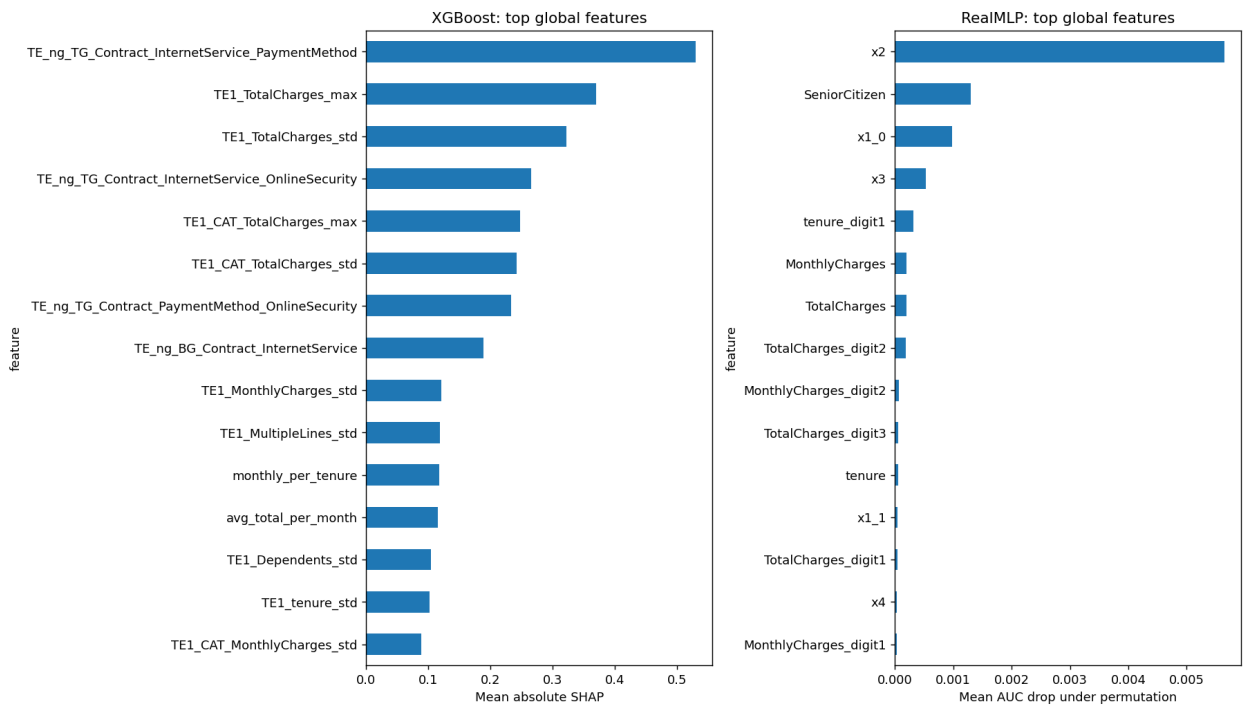
Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

5. Global feature reliance: ranking comparison

```
xgb_top = xgb_mean_abs_shap.sort_values("mean_abs_shap",
ascending=False).head(15).copy()
real_top = real_perm.sort_values("mean_importance",
ascending=False).head(15).copy()

fig, axes = plt.subplots(1, 2, figsize=(14, 8))

xgb_top.sort_values("mean_abs_shap").plot(
    x="feature", y="mean_abs_shap", kind="barh", legend=False,
ax=axes[0]
)
axes[0].set_title("XGBoost: top global features")
axes[0].set_xlabel("Mean absolute SHAP")
```

```
real_top.sort_values("mean_importance").plot(
    x="feature", y="mean_importance", kind="barh", legend=False,
ax=axes[1]
)
axes[1].set_title("RealMLP: top global features")
axes[1].set_xlabel("Mean AUC drop under permutation")

plt.tight_layout()
plt.show()

display(pd.DataFrame({
    "xgb_top_features": xgb_top["feature"].tolist(),
    "realmlp_top_features": real_top["feature"].tolist(),
}))
```




```
                                      xgb_top_features
realmlp_top_features
0    TE_ng_TG_Contract_InternetService_PaymentMethod
x2
1                                TE1_TotalCharges_max
SeniorCitizen
2                                TE1_TotalCharges_std
x1_0
3   TE_ng_TG_Contract_InternetService_OnlineSecurity
x3
4                            TE1_CAT_TotalCharges_max
tenure_digit1
```

```
5                              TE1_CAT_TotalCharges_std
MonthlyCharges
6     TE_ng_TG_Contract_PaymentMethod_OnlineSecurity
TotalCharges
7                  TE_ng_BG_Contract_InternetService
TotalCharges_digit2
8                              TE1_MonthlyCharges_std
MonthlyCharges_digit2
9                               TE1_MultipleLines_std
TotalCharges_digit3
10                                monthly_per_tenure
tenure
11                               avg_total_per_month
x1_1
12                                TE1_Dependents_std
TotalCharges_digit1
13                                    TE1_tenure_std
x4
14                         TE1_CAT_MonthlyCharges_std
MonthlyCharges_digit1
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is a cross-model comparison, but the scales are not the same, so compare **rankings and concentration**, not absolute magnitudes. If the tree model is dominated by a small number of engineered variables while the neural model spreads reliance across more inputs, that supports the claim that the two families exploit the feature space differently.

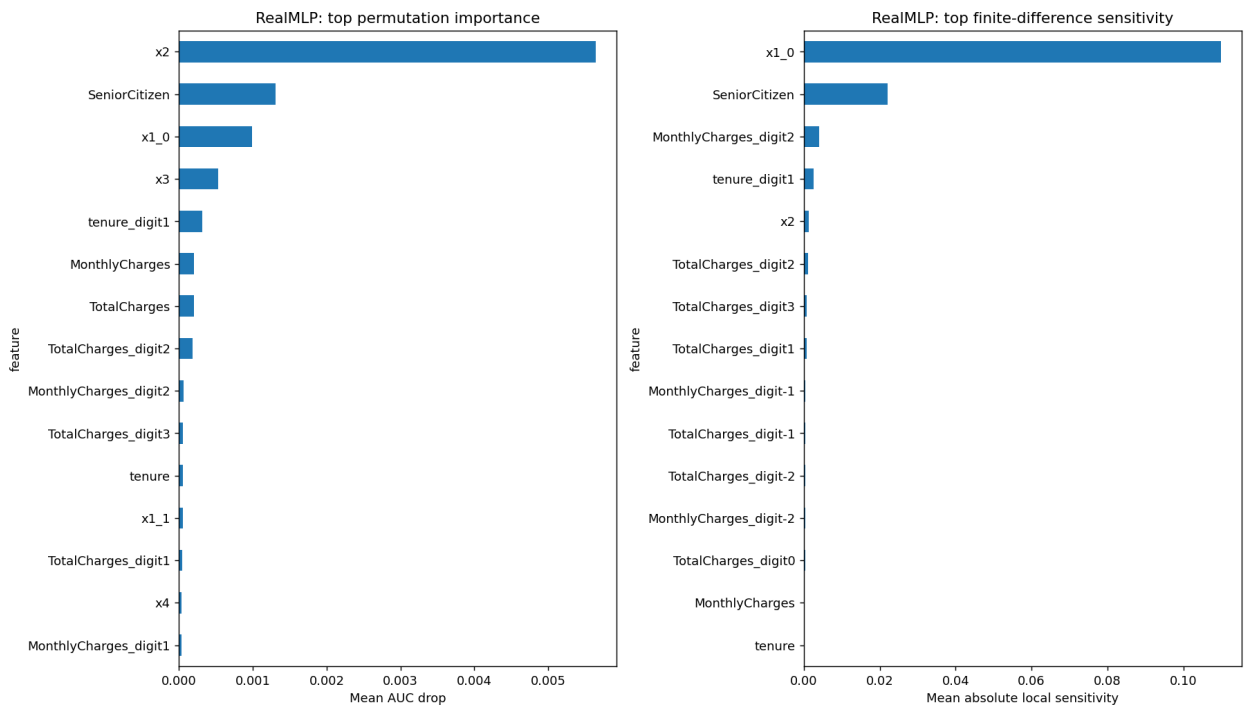
Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

6. RealMLP-only internal sensitivity diagnostics

```
fig, axes = plt.subplots(1, 2, figsize=(14, 8))

real_perm.head(15).sort_values("mean_importance").plot(
    x="feature", y="mean_importance", kind="barh", legend=False,
ax=axes[0]
)
```

```
axes[0].set_title("RealMLP: top permutation importance")
axes[0].set_xlabel("Mean AUC drop")

real_sens.head(15).sort_values("mean_abs_sensitivity").plot(
    x="feature", y="mean_abs_sensitivity", kind="barh", legend=False,
ax=axes[1]
)
axes[1].set_title("RealMLP: top finite-difference sensitivity")
axes[1].set_xlabel("Mean absolute local sensitivity")

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is a **within-RealMLP** analysis and should be interpreted in isolation, not numerically compared against XGBoost. Compare the two panels to see whether the features that matter most globally are also the most locally responsive. Large permutation importance with modest sensitivity can indicate a feature that matters through broader structural effects rather than steep local gradients.

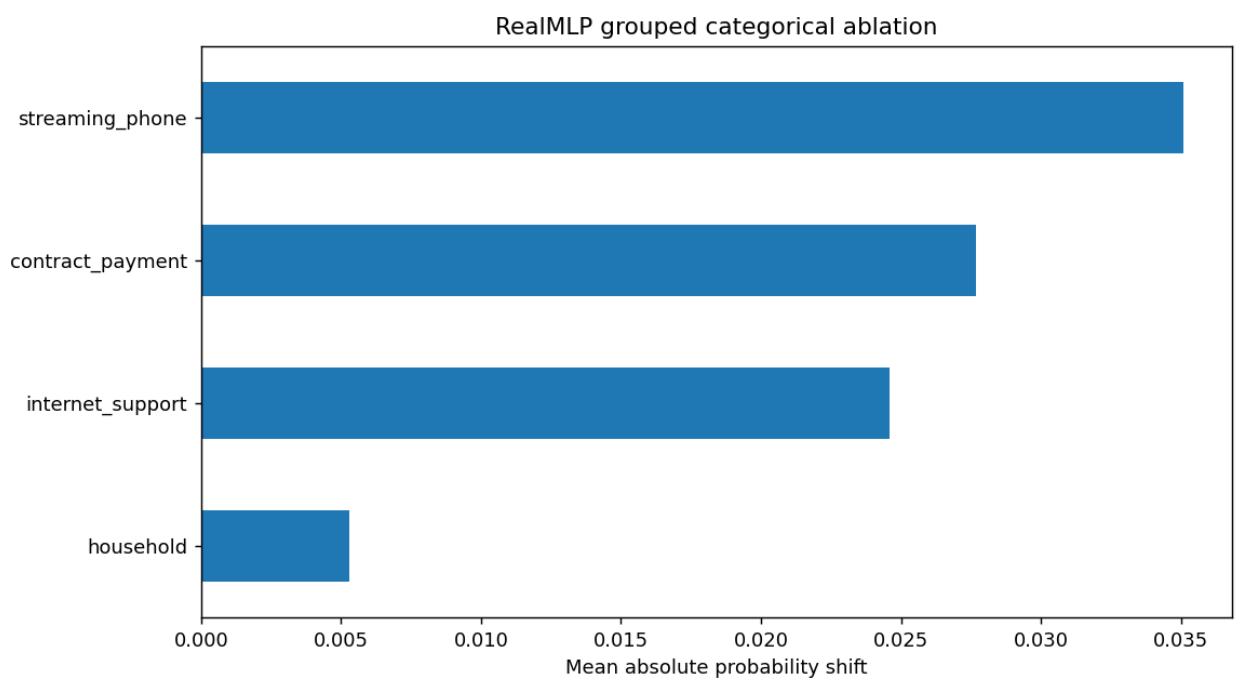
Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

7. RealMLP-only grouped ablation

```
display(real_ablation)

ax = real_ablation.sort_values("mean_abs_delta").plot(
    x="group",
    y="mean_abs_delta",
    kind="barh",
    legend=False,
    figsize=(9, 5),
)
ax.set_title("RealMLP grouped categorical ablation")
ax.set_xlabel("Mean absolute probability shift")
ax.set_ylabel("")
plt.tight_layout()
plt.show()
```

```
              group  mean_abs_delta  mean_signed_delta  max_abs_delta
0   streaming_phone        0.035071           0.034967       0.223998
1  contract_payment        0.027666          -0.027666       0.198302
2  internet_support        0.024606          -0.024606       0.317824
3         household        0.005301          -0.004702       0.110057
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

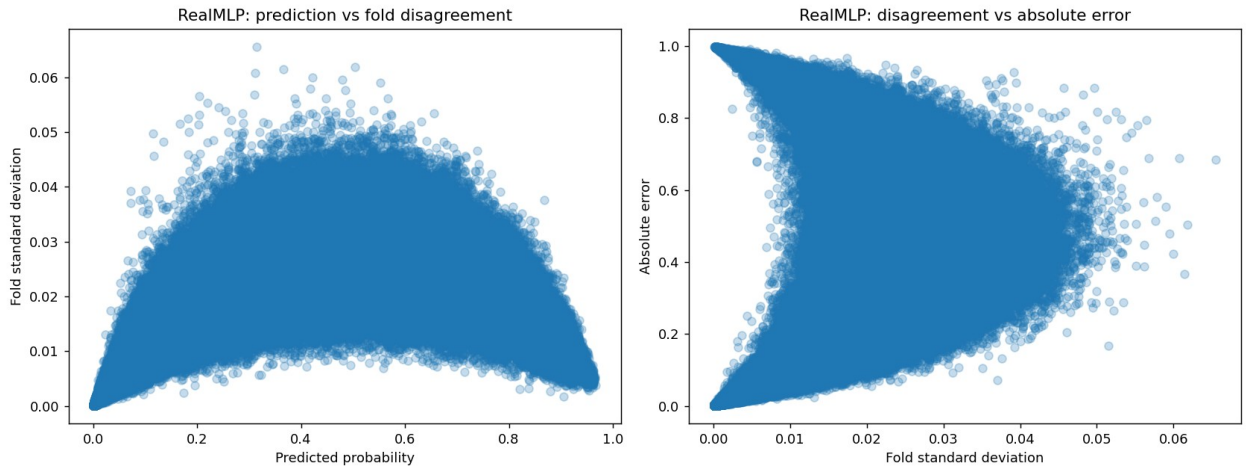
This is also a **within-RealMLP** test. It tells you which semantic subsystems the network uses most strongly. Use it to say whether the network is driven more by contractual/payment structure, service bundle structure, or demographic structure. Do not compare the raw magnitudes here against SHAP values from XGBoost.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

8. RealMLP uncertainty and disagreement

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].scatter(real_train["realmlp_pred"], real_train["fold_std"],
alpha=0.25)
axes[0].set_title("RealMLP: prediction vs fold disagreement")
axes[0].set_xlabel("Predicted probability")
axes[0].set_ylabel("Fold standard deviation")

axes[1].scatter(real_train["fold_std"], real_train["abs_error"],
alpha=0.25)
axes[1].set_title("RealMLP: disagreement vs absolute error")
axes[1].set_xlabel("Fold standard deviation")
axes[1].set_ylabel("Absolute error")

plt.tight_layout()
plt.show()

display(real_unc)
```




```
                          mean_pred  observed_rate  mean_abs_error
n
fold_std

(-0.0009382, 0.000666]   0.003392       0.001750        0.005128
118839
(0.000666, 0.00317]      0.012918       0.011335        0.023767
118839
(0.00317, 0.0136]        0.147528       0.151021        0.138210
118838
(0.0136, 0.0233]         0.411039       0.442700        0.354451
118839
(0.0233, 0.0656]         0.474269       0.519232        0.449057
118839
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

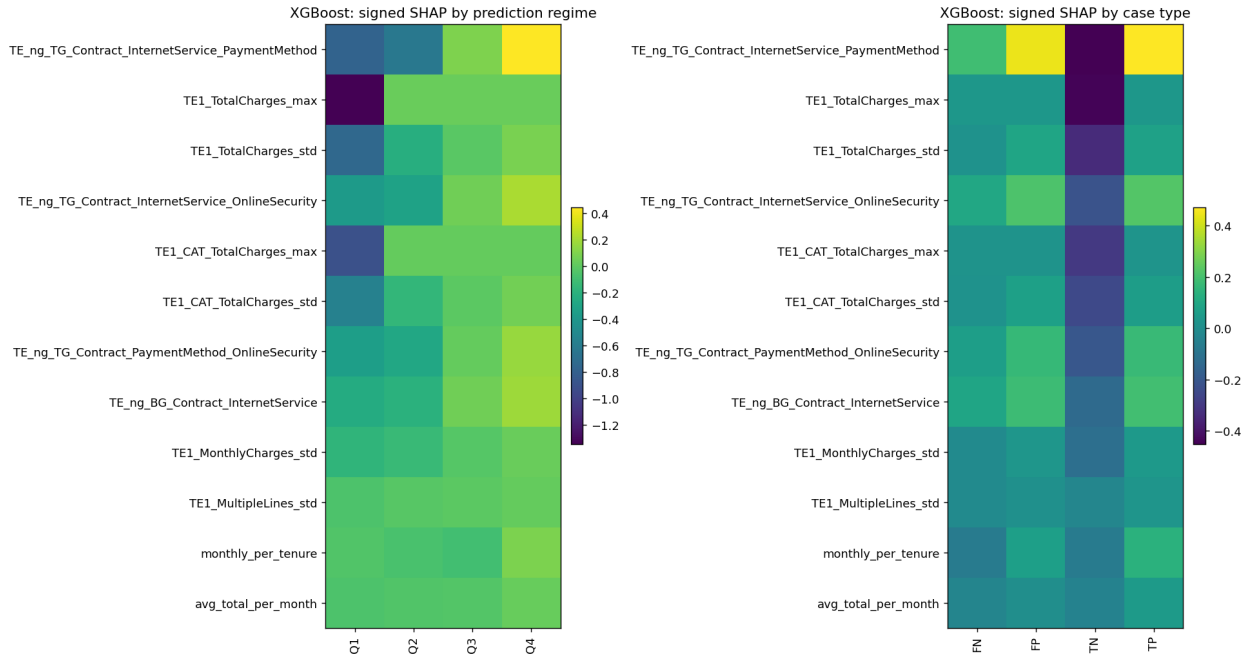
This is a **within-RealMLP** uncertainty analysis. In the left panel, look for whether disagreement is concentrated near the middle of the score range rather than the extremes. In the right panel, look for whether larger disagreement tends to coincide with larger error. If it does, then fold dispersion is acting as a meaningful uncertainty proxy rather than just random variation.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

9. XGBoost regime and case-type explanation structure

```
# choose a compact set of influential XGB features for visual
comparison
top_xgb_features = xgb_mean_abs_shap.sort_values("mean_abs_shap",
ascending=False)["feature"].head(12).tolist()

fig, axes = plt.subplots(1, 2, figsize=(15, 8))

regime_plot = xgb_regime_signed[top_xgb_features] if
set(top_xgb_features).issubset(xgb_regime_signed.columns) else
xgb_regime_signed.iloc[:, :12]
im0 = axes[0].imshow(regime_plot.values, aspect="auto")
axes[0].set_title("XGBoost: signed SHAP by prediction regime")
axes[0].set_yticks(range(len(regime_plot.index)))
axes[0].set_yticklabels(regime_plot.index)
axes[0].set_xticks(range(len(regime_plot.columns)))
axes[0].set_xticklabels(regime_plot.columns, rotation=90)
plt.colorbar(im0, ax=axes[0], fraction=0.046, pad=0.04)

case_plot = xgb_case_type_signed[top_xgb_features] if
set(top_xgb_features).issubset(xgb_case_type_signed.columns) else
xgb_case_type_signed.iloc[:, :12]
im1 = axes[1].imshow(case_plot.values, aspect="auto")
axes[1].set_title("XGBoost: signed SHAP by case type")
axes[1].set_yticks(range(len(case_plot.index)))
axes[1].set_yticklabels(case_plot.index)
axes[1].set_xticks(range(len(case_plot.columns)))
axes[1].set_xticklabels(case_plot.columns, rotation=90)
plt.colorbar(im1, ax=axes[1], fraction=0.046, pad=0.04)

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is an **XGBoost-only** interpretability section. It should be taken in isolation. The left panel asks how the signed contribution of important features changes as you move from low-risk to high-risk score regimes. The right panel asks whether the model is using the same features differently in true positives, false positives, true negatives, and false negatives. This is where you diagnose failure modes in the tree model rather than compare absolute attribution values against the neural model.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

10. Previously generated model-specific figures for the appendix or report

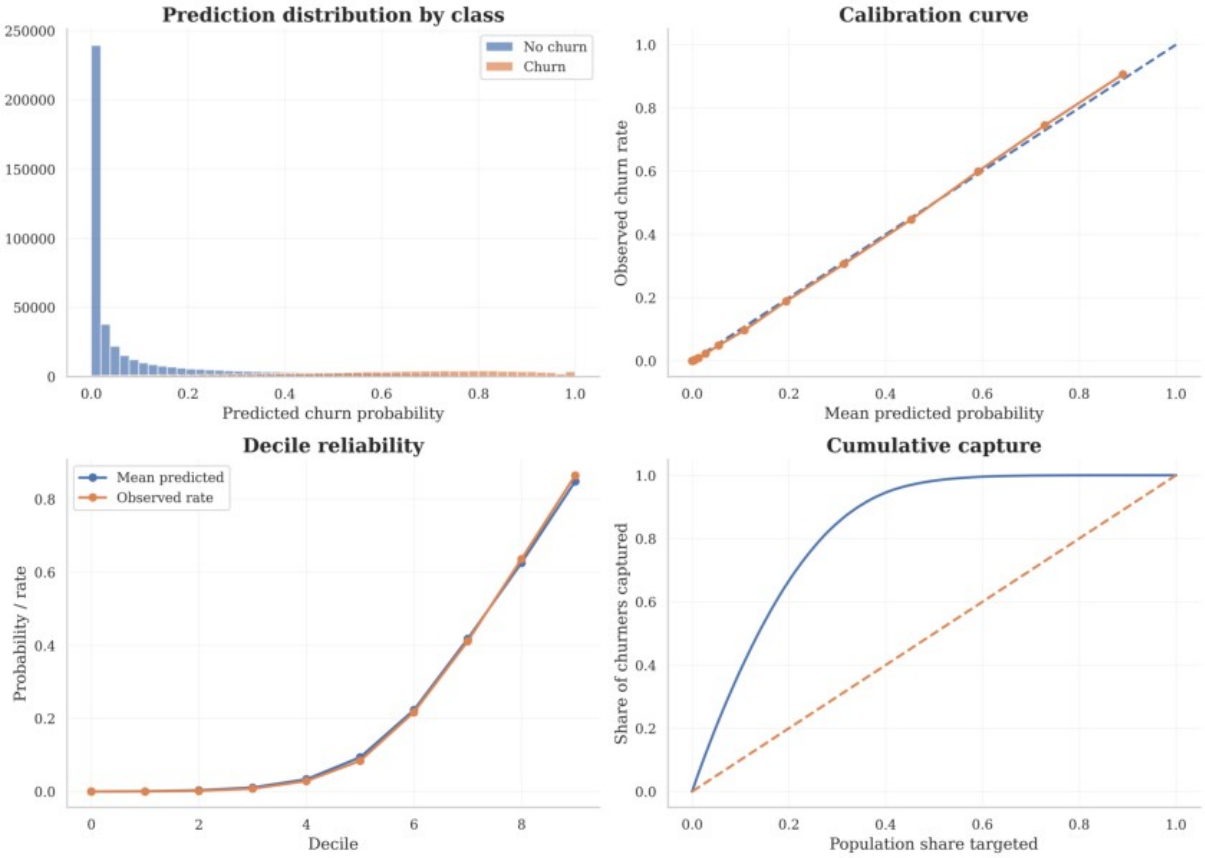
```
existing_figures = [
    "xgb_01_core_diagnostics.png",
    "xgb_02_mean_abs_shap.png",
    "xgb_03_shap_beeswarm.png",
```

```
    "xgb_04_regime_signed_shap_heatmap.png",
    "xgb_05_signed_shap_by_case_type.png",
    "realmlp_01_core_diagnostics.png",
    "realmlp_02_permutation_importance.png",
    "realmlp_03_finite_difference_sensitivity.png",
    "realmlp_04_grouped_ablation.png",
    "realmlp_05_disagreement_vs_error.png",
]

available_figures = [f for f in existing_figures if (OUTDIR /
f).exists()]
print("Available pre-rendered figures:")
for f in available_figures:
    print("-", f)
```

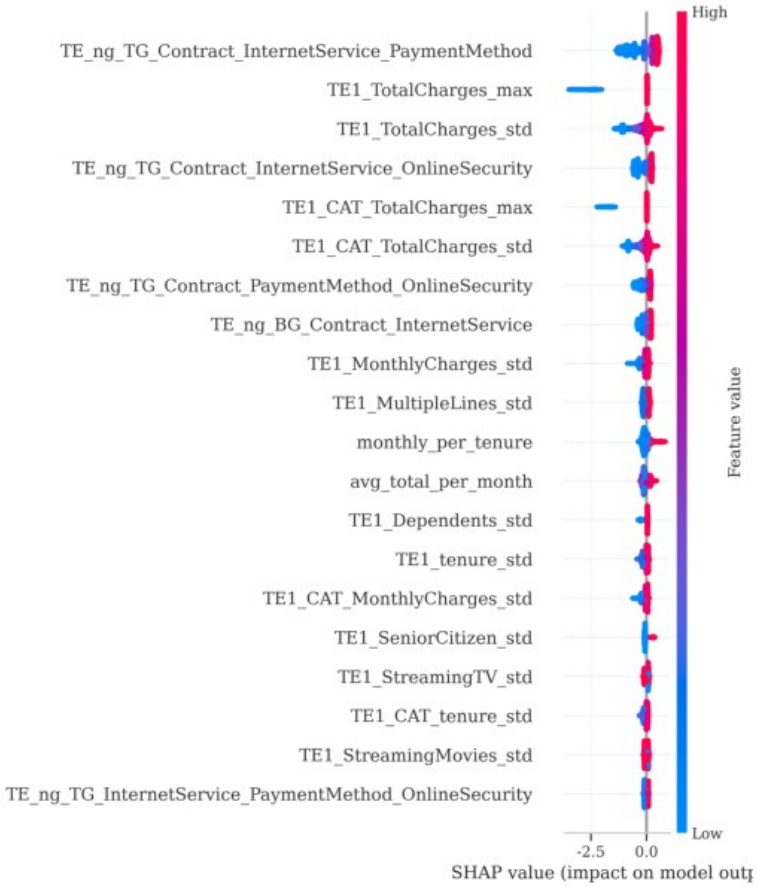
```
Available pre-rendered figures:
- xgb_01_core_diagnostics.png
- xgb_02_mean_abs_shap.png
- xgb_03_shap_beeswarm.png
- xgb_04_regime_signed_shap_heatmap.png
- xgb_05_signed_shap_by_case_type.png
- realmlp_01_core_diagnostics.png
- realmlp_02_permutation_importance.png
- realmlp_03_finite_difference_sensitivity.png
- realmlp_04_grouped_ablation.png
- realmlp_05_disagreement_vs_error.png
```

```
# Display a small curated subset without recreating them.
to_show = [
    "xgb_01_core_diagnostics.png",
    "xgb_03_shap_beeswarm.png",
    "realmlp_01_core_diagnostics.png",
    "realmlp_05_disagreement_vs_error.png",
]

for fname in to_show:
    fpath = OUTDIR / fname
    if fpath.exists():
        img = mpimg.imread(fpath)
        plt.figure(figsize=(10, 6))
        plt.imshow(img)
        plt.axis("off")
        plt.title(fname)
        plt.show()
```

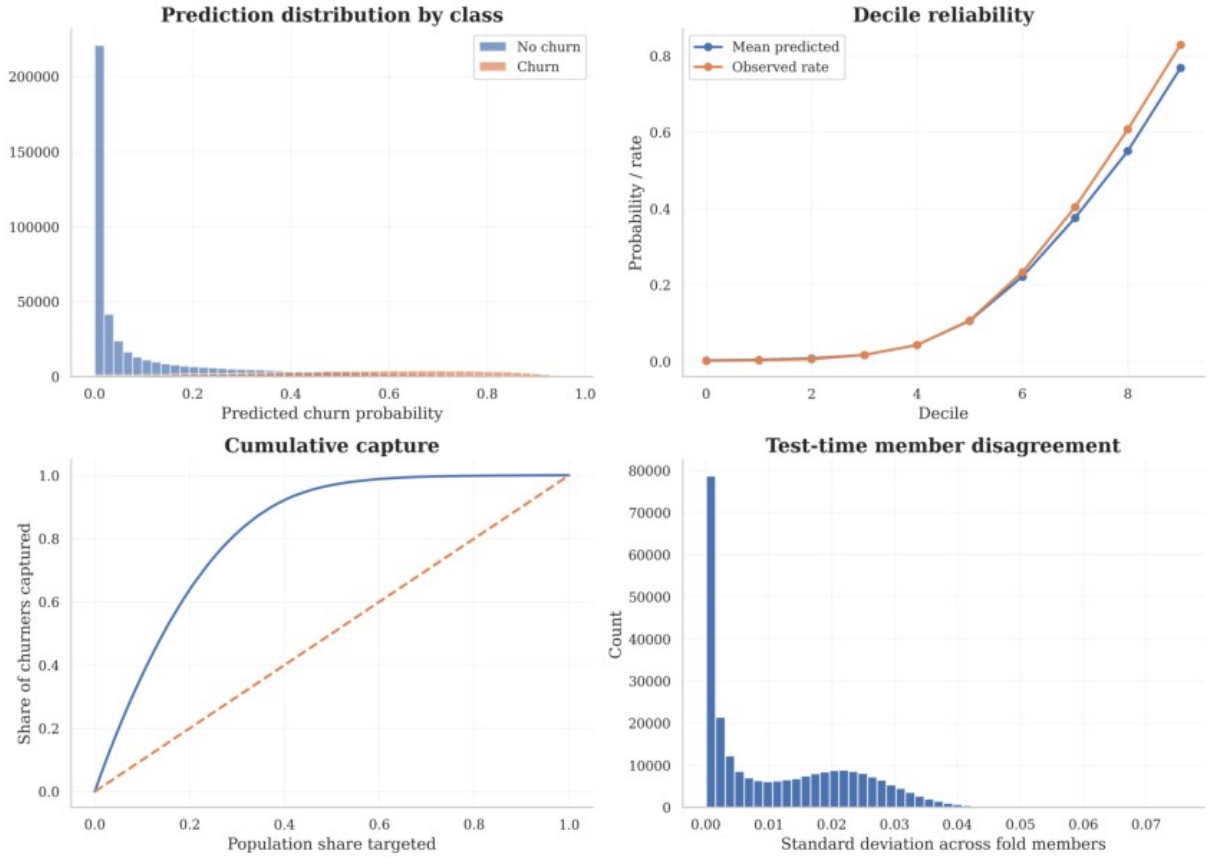
xgb_01_core_diagnostics.png



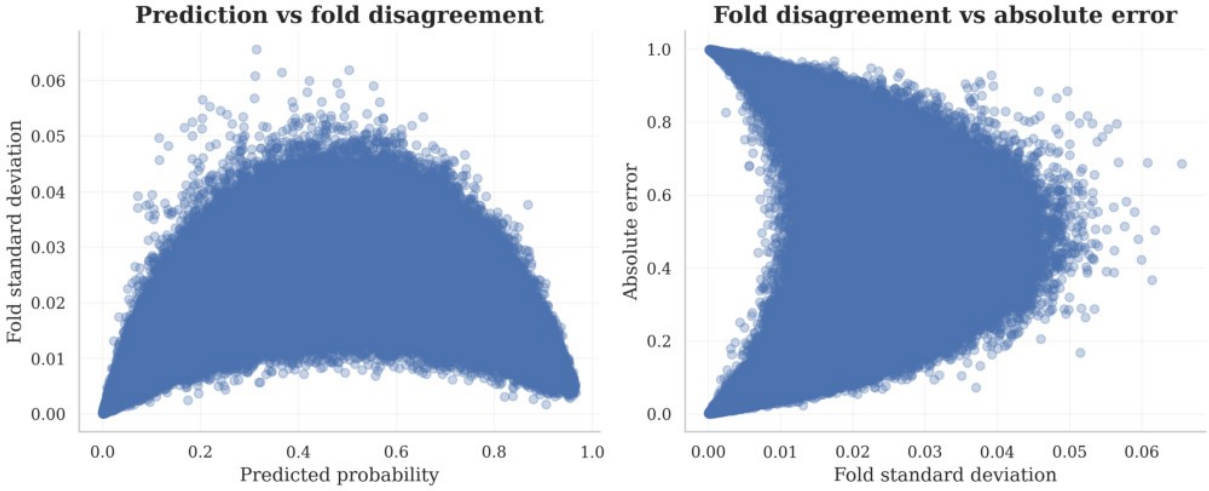
xgb_03_shap_beeswarm.png



realmlp_01_core_diagnostics.png



realmlp_05_disagreement_vs_error.png



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are the original notebook figures, kept for direct reuse in the report appendix. They are best used as supporting material after the synthesis plots above. In the report body, lead with the comparison tables and comparison plots first, then use these as model-specific evidence.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

11. Final report-ready tables

```
# concise report tables
report_table_metrics = comparison_summary.copy()

report_table_top_features = pd.DataFrame({
    "XGBoost top SHAP features":
xgb_mean_abs_shap.sort_values("mean_abs_shap", ascending=False)
["feature"].head(10).tolist(),
    "RealMLP top permutation features":
real_perm.sort_values("mean_importance", ascending=False)
["feature"].head(10).tolist(),
})

# risk concentration summary
report_table_risk = pd.DataFrame([
    {
        "model": "XGBoost",
        "lowest_decile_observed_rate":
dec_compare["xgb_observed_rate"].iloc[0],
        "highest_decile_observed_rate":
dec_compare["xgb_observed_rate"].iloc[-1],
        "spread": dec_compare["xgb_observed_rate"].iloc[-1] -
dec_compare["xgb_observed_rate"].iloc[0],
    },
    {
        "model": "RealMLP",
        "lowest_decile_observed_rate":
dec_compare["real_observed_rate"].iloc[0],
        "highest_decile_observed_rate":
dec_compare["real_observed_rate"].iloc[-1],
```

```
        "spread": dec_compare["real_observed_rate"].iloc[-1] -
dec_compare["real_observed_rate"].iloc[0],
    },
])

display(report_table_metrics)
display(report_table_risk)
display(report_table_top_features)
```

```
     model       auc   logloss     brier  n_features_runtime  \
0  XGBoost  0.933437  0.268055  0.085993                 161
1  RealMLP  0.917199  0.298696  0.095583                  22

   n_feature_importance_table_rows
0                              161
1                               22

     model  lowest_decile_observed_rate  highest_decile_observed_rate
\
0  XGBoost                     0.000000                      0.865012

1  RealMLP                     0.001094                      0.829670


     spread
0  0.865012
1  0.828576

                          XGBoost top SHAP features  \
0   TE_ng_TG_Contract_InternetService_PaymentMethod
1                             TE1_TotalCharges_max
2                             TE1_TotalCharges_std
3  TE_ng_TG_Contract_InternetService_OnlineSecurity
4                        TE1_CAT_TotalCharges_max
5                        TE1_CAT_TotalCharges_std
6    TE_ng_TG_Contract_PaymentMethod_OnlineSecurity
7                 TE_ng_BG_Contract_InternetService
8                           TE1_MonthlyCharges_std
9                            TE1_MultipleLines_std

  RealMLP top permutation features
0                               x2
1                    SeniorCitizen
2                             x1_0
3                               x3
4                    tenure_digit1
5                   MonthlyCharges
6                     TotalCharges
7               TotalCharges_digit2
8             MonthlyCharges_digit2
9               TotalCharges_digit3
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are the tables to cite directly in a short report. The first gives the predictive headline numbers. The second gives the segmentation story. The third gives the interpretability headline: what each model appears to care about most.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

12. Optional export of report tables

```
REPORT_DIR = OUTDIR / "report_tables"
REPORT_DIR.mkdir(exist_ok=True)

report_table_metrics.to_csv(REPORT_DIR / "report_metrics.csv",
index=False)
report_table_risk.to_csv(REPORT_DIR / "report_risk_concentration.csv",
index=False)
report_table_top_features.to_csv(REPORT_DIR /
"report_top_features.csv", index=False)
dec_compare.to_csv(REPORT_DIR / "report_decile_comparison.csv",
index=False)

print("Saved report tables to:", REPORT_DIR.resolve())
```

```
Saved report tables to: /Users/abhishekdey/Downloads/Predicting-
Churn/analysis_outputs/report_tables
```